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Project Title:	Dynamics and Structure in Colloidal Suspensions Far from Equilibrium		
Division Name:	Physical Sciences		
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A dense pack of solid particles can be fluidized by an upward flow of fluid, which counterbalances the gravitational force on the particles. A liquid-fluidized bed of small particles (of the order of 10-100 microns) can be prepared in a uniformly fluidized state; i.e. one without large-scale density fluctuations. In the case of gas-fluidized beds however, numerous studies have found that the particles are unstable to gas bubble formation just at the onset of uniform fluidization. An understanding of the dynamics of gas-fluidized beds raises interesting and important scientific issues: What is the role of inertia? What are the limits of stability? What are the characteristic patterns of the unstable flows? The primary difficulty in studying gas-fluidized beds experimentally is that the buoyancy mismatch makes the gravitational force on the particles unavoidably large. These systems can only be fluidized by high-velocity gas flows where the dynamics is immediately very complicated, and the bed is unstable as soon as it is fluidized. Thus it is not known whether or not gas-fluidized beds are stable to uniform fluidization at sufficiently low Stokes number, as would

Task Description:	be expected from purely fluid-dynamical considerations, or whether other forces, such as cohesion, play a destabilizing role. These critical questions can not be addressed because stable gas fluidization can not be achieved in earth's gravity. In order to study the onset of a particle-induced instability, the minimum fluidization velocity must be reduced by several orders of magnitude. This can only accomplished in a microgravity environment. In our current NASA sponsored research we have focused on the sedimentation of monodisperse liquid-particle suspensions. Numerical simulations have led to a breakthrough in understanding the origin of a correlation length that appears spontaneously in a settling suspension. Nevertheless laboratory experiments have raised doubts that our understanding of the dynamics at low Reynolds number is complete. In particular, the experiments raise issues of the effects of system size, initial conditions, and polydispersity, which have been largely ignored up to now. Our new proposal will build on these prior accomplishments. Its overall objective is to continue our ongoing efforts to develop a better understanding of gas fluidized systems by extending the current limits of laboratory experiments and numerical simulation. We use experiments and simulations synergistically; thus we not only use laboratory experiments to validate the results of numerical simulations, but we also use simulations to clarify aspects of laboratory experiments, such as polydispersity, that are difficult to control experimentally. Specifically we will attempt to answer three fundamental questions related to particle-fluid flows, motivated in part by what was discovered in the past 3 years research. The first question is to understand the origin of the observed correlation length in settling suspensions and decide if it is universal, applying also to fluidized suspensions. The second issue is to discover if particle collisions in a low-Reynolds number fluidized bed lead to important changes in suspension microstructure at finite Stokes numbers, and to investigate whether such microstructural changes are related to the onset of instability in gas-fluidized suspensions at low Stokes numbers. Finally we want to examine a flow with a non-uniform external force, namely a rotating suspension flow. In this case the centrifugal force produces a non-uniform body force on the particles, leading to a surprising segregation of particles along the rotation axis into bands of high and low concentration. The bands of high and low particle concentration may develop differently in a microgravity environment than on earth. We propose a coordinated program of numerical simulations and laboratory experiments to accomplish the proposal objectives. Numerical simulations will be used to probe parameter regimes that are unavailable to laboratory experiments, and also to assess t [abstract truncated]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We found that much of the results we were obtaining were highly susceptible to the initial conditions, which depended very delicately on the pre-mix. This lead to confusing and somewhat irreproducible results. To overcome these limitations, we built a fluidized bed which achieves the settling by flowing water up through the particles at the same rate they are settling, to achieve a steady state condition. This avoids the problems with the initial conditions as we can wait until the fluidized bed stabilizes in a steady state. However, this does lead to somewhat different conditions since the time evolution of the settling is modified. We have successfully studied the fluidize bed behavior and are currently in the process of analyzing the results and preparing some publications to summarize them.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Mucha PJ, Tee S-Y, Weitz DA, Shraiman BI, Brenner MP. "A model for velocity fluctuations in sedimentation." Journal of Fluid Mechanics, 2004 Feb;501:71-104. http://dx.doi.org/10.1017/S0022112003006967 , Feb-2004